BCA 1st Semester - Practical Lab Assignments Guide

Assignment 3: Build Your Own LAN + Become IT Support

Objective

After completing this assignment, you should be able to:

- Create a small network using a mobile hotspot (LAN)
- Find the IP address of a Windows device
- Understand how Local Area Networks (LANs) work
- Test connectivity using Windows ping command
- Share files between Windows computers
- Understand internet speed and latency (ping time)
- Perform basic Windows internet troubleshooting

Individual Assignment (Solo work - NOT a group project)

What You Need:

- 1 Windows Laptop (your own)
- 1 Mobile Hotspot (your own phone)
- 1 Additional Device (smartphone or tablet - can be yours or borrowed)
- No network cables required!
  Note for Students with Limited Devices: If you don't have access to multiple devices, you can still complete this assignment by:
  - Using your laptop and phone as two networked devices
  - Demonstrating hotspot connection and basic PING commands
  - Testing file sharing concepts with what you have available
  - Your teacher understands device limitations and will grade based on what you can accomplish

Total Marks: 15 points

- Create LAN + file sharing (6 marks)
- Speed vs latency measurements (4 marks)
- Troubleshooting card (5 marks)

Task 1: Create and Test Your LAN (6 marks)

Step 1: Create the Mobile Hotspot

On Your Phone:

1. Open **Settings**
2. Find **Mobile Hotspot** or **Personal Hotspot**
3. Turn it **ON**
4. Set a network name (SSID) like: **BCA_LAN_BCA001**
5. Set a password (write it down!)
6. Note the password somewhere

On Your Laptop & Additional Device:

1. Open **Wi-Fi Networks** (bottom right of Windows taskbar)
2. Click the network name **BCA_LAN_BCA001**
3. Click **Connect**
4. Enter the password
5. Wait for connection

✓ **Congratulations!** You have created a LAN!

Step 2: Find IP Addresses

Every device on a network has a unique IP address (like a house address).

On Your Laptop:

1. Open **Command Prompt** (Press Windows Key + type "cmd")
2. Type: `ipconfig`
3. Press **Enter**
4. Look for "IPv4 Address" (example: 192.168.43.10)
5. Write it down: _____

On Your Additional Device:

1. Repeat the same steps
2. Write down its IP: _____

On Your Phone/Tablet:

1. Open **Settings**
2. Find **Network & Internet** or **Wi-Fi**
3. Click on the connected network
4. Look for "IP Address"
5. Write it down: _____

Important Question: Do all three devices have similar IP addresses? (Example: All start with 192.168.43...) Answer: Yes / No

If YES, you're on the same LAN!

Step 3: Test Connection with PING

What is PING?

PING is a Windows command that tests if another device is reachable on your network.

On Laptop A:

1. Open Command Prompt
2. Type: `ping 192.168.43.20` (use your additional device's IP address)
3. Press **Enter**
4. Watch the output

You should see:

```
Reply from 192.168.43.20: bytes=32 time=5ms TTL=64
Reply from 192.168.43.20: bytes=32 time=4ms TTL=64
Reply from 192.168.43.20: bytes=32 time=6ms TTL=64
Reply from 192.168.43.20: bytes=32 time=5ms TTL=64
```

This means: **Connection is working!** ✓

If you see:

```
Request timed out
Request timed out
```

This means: **Connection failed.** Check if both devices are on the same hotspot.

Take a Screenshot:

- Show the successful PING result
- Include Name + Roll No. in corner
- Save as: Screenshot_PING_Success.png

Step 4: Share Files Between Laptops

Now you'll transfer files without using the internet!

On Your Laptop (Share Creator):

1. Create a folder on Desktop called: **BCA_Shared**
2. Add some files (documents, images, etc.)
3. Right-click the folder → **Properties**
4. Click **Sharing** tab
5. Click **Share...**
6. Select **Everyone**
7. Set Permission: **Read**
8. Click **Share** and note the network path

On Your Additional Device (Access the Share):

1. Open **File Explorer** (Windows Key + E)
2. In the address bar, type: `\\192.168.43.10\BCA_Shared` (use your laptop's IP)
3. Press **Enter**
4. You should see the shared files!
5. Try opening a file

Important Question: Did you need to buy an internet recharge to share files? Why?

Answer:

_____

Hint: Data travels within your local network, not over the internet.

Take a Screenshot:

- Show your additional device accessing the shared folder
- Include Name + Roll No.
- Save as: Screenshot_FileSharing.png

Task 2: Internet Speed vs Latency (4 marks)

What's the Difference?

Speed (Bandwidth):

- How much data can download at once
- Measured in Mbps (Megabits per second)
- Think: How thick is the water pipe?
- Affects: Downloading, streaming video quality

Latency (Ping):

- How long it takes for data to reach you

- Measured in ms (milliseconds)
- Think: How far away is the water source?
- Affects: Video calls, online gaming

Step 1: Measure Internet Speed

1. Open a web browser
2. Go to: **www.fast.com**
3. Wait for the test to complete
4. Write down your speed: _____ Mbps

Step 2: Measure Latency (Ping)

1. Open Command Prompt
2. Type: `ping google.com -n 10`
3. Press **Enter**
4. Look at the end for: "Average = ___ ms"
5. Write down: _____ ms

Quality Interpretation:

- 0-20 ms: Excellent (gaming ready!)
- 20-50 ms: Good
- 50-100 ms: Fair
- 100+ ms: Poor (noticeable lag)

Step 3: Test Impact of Streaming

Now let's see how YouTube affects your ping.

Before YouTube:

1. Run ping again: `ping google.com -n 10`
2. Write average ping: _____ ms

While YouTube is playing:

1. On your additional device (phone/tablet), open YouTube

2. Start playing any video
3. On a laptop, run: `ping google.com -n 10` (while video plays)
4. Write average ping: _____ ms

Question: What changed?

Answer:

_____

Expected Answer: Latency probably increased because YouTube was using bandwidth, causing other data to wait longer.

Step 4: Why Latency Matters for Gaming

In Online Games:

- Low latency (5-20 ms) = Your actions happen instantly ✓
- High latency (100+ ms) = Lag, everything feels slow ✗

In Video Calls:

- Low latency = Conversation flows naturally ✓
- High latency = People talk over each other ✗

Question: Why is low latency important for online gaming and video calls?

Answer:

_____

Take Screenshots:

- Screenshot of fast.com result
- Screenshot of ping results (before YouTube)
- Screenshot of ping results (during YouTube)
- All with Name + Roll No. visible

Task 3: Become an IT Support Person (5 marks)

Create a Troubleshooting Card

You'll create a quick reference guide for fixing internet problems on Windows.

What to Create:

- 1 A4 sheet (white paper)
- Handwritten or printed
- Title: **WINDOWS INTERNET NOT WORKING? DO THIS IN ORDER**
- 6 troubleshooting steps with boxes/arrows
- Make it colorful and easy to follow

6-Step Troubleshooting Process

Step 1: Check Wi-Fi Symbol

- Look at Windows taskbar (bottom right)
- Do you see a Wi-Fi icon?
- Is it connected or showing "No internet"?
- If not connected, click and reconnect

Step 2: Check IP Configuration

- Open Command Prompt
- Type: `ipconfig`
- Look for IPv4 Address
- Should start with 192.168.x.x (not 169.x.x.x)
- If 169.x.x.x, DHCP is broken

Step 3: Ping Your Gateway

- In ipconfig output, find: "Default Gateway" (usually 192.168.1.1)
- Type: `ping 192.168.1.1`
- If it works: Your local network is OK

- If it fails: Your gateway/router is down

Step 4: Test Internet Connection

- Type: `ping 8.8.8.8` (Google's Public DNS)
- If it works: You have internet!
- If it fails: Internet is down

Step 5: Test Domain Names

- Type: `ping google.com`
- If it works: DNS is working
- If it fails but 8.8.8.8 worked: DNS is broken

Step 6: Restart and Reconnect

- Disconnect from Wi-Fi
- Wait 5 seconds
- Reconnect to the same Wi-Fi
- If that doesn't work: Restart Windows
- If still broken: Restart your router

Card Design Tips

Make it attractive:

- Use different colors for each step
- Draw simple icons/symbols
- Use arrows to show flow ↓
- Make text large and readable
- Use boxes or circles for each step

Example Layout:

```
┌─────────────────────────────────────┐
│ Wi-Fi Not Working?                  │
│ DO THIS IN ORDER                    │
```

```
└──────────┬──────────────────────────┘
           ↓
    [1. Check Wi-Fi]
           ↓
   [2. Run ipconfig]
           ↓
  [3. Ping Gateway]
           ↓
   [4. Ping 8.8.8.8]
           ↓
  [5. Ping google.com]
           ↓
 [6. Restart/Reconnect]
```

Take a Photo:

- Clear, well-lit photo of your card
- Include Name + Roll No. on the card or nearby
- Save as: Photo_TroubleshootingCard.png

Assignment 2 Submission Checklist

- ☐ Your mobile hotspot created and activated
- ☐ Your laptop connected to hotspot
- ☐ Your additional device connected to hotspot
- ☐ (Optional) Additional device(s) connected
- ☐ IP addresses recorded for your devices
- ☐ Same-subnet question answered
- ☐ PING test successful between your devices
- ☐ Shared folder created on your laptop
- ☐ File sharing tested from your additional device

- ☐ Internet speed test completed (fast.com)
- ☐ Ping test completed (normal, with YouTube)
- ☐ Latency observations recorded
- ☐ Gaming/video call latency question answered
- ☐ Troubleshooting card created (A4 handwritten)
- ☐ All screenshots and photos taken
- ☐ Your Name + Roll No. visible in all screenshots/photos

Submission

File name format: RollNo_Name_Assignment2.pdf

Example: BCA001_Rahul_Kumar_Assignment2.pdf

Include in PDF:

- All screenshots (3-4 PING screenshots)
- Screenshot of your connected devices on hotspot
- Photo of troubleshooting card
- Written answers to all questions
- 3-4 pages total

Important Notes for Both Assignments

⚠ DO NOT COPY

- Each student's computer is different
- Your CPU, RAM, IP address, ping time will be DIFFERENT from friends
- **Different answers are EXPECTED**
- This proves you did your own work!
- If everyone has the same answers: Academic dishonesty suspected

Screenshot Requirements

- Every screenshot must show Name + Roll No.
- Use Notepad with your information visible
- Or write your Roll No. on a chit and show it
- Clear, readable text
- Good lighting and focus

Windows Commands Summary

Command	What it does	Example
whoami	Shows your username	COMPUTER\YourName
ipconfig	Shows network info	IPv4: 192.168.1.5
dir	Lists files/folders	Desktop, Documents
mkdir BCA_Test	Creates new folder	Creates "BCA_Test"
tasklist	Lists running programs	explorer.exe, chrome.exe
systeminfo	Shows Windows info	Processor, RAM, OS
ping 8.8.8.8	Tests internet	Reply from 8.8.8.8
ping google.com	Tests DNS	Reply from google.com

Quick Glossary

- **CPU:** Processor, the "brain" of your computer
- **RAM:** Memory, temporary storage for running programs
- **SSD:** Fast solid-state drive, no moving parts
- **HDD:** Slow hard disk drive, has spinning disk
- **GPU:** Graphics card, handles video/gaming

- **IP Address:** Network address, like a house address
- **LAN:** Local Area Network, computers connected together
- **PING:** Test command to check connectivity
- **Latency:** Delay for data to travel (ms)
- **Speed/Bandwidth:** How much data per second (Mbps)
- **UAC:** User Account Control, Windows permission dialog
- **DHCP:** Automatic IP address assignment
- **Gateway:** Device connecting to internet (router)
- **DNS:** Converts google.com to IP address
- **Command Prompt:** Text-based Windows interface
- **Task Manager:** Windows system information tool

Final Tips for Success

✓ **Do the practical work yourself** - Don't just copy screenshots

✓ **Different answers = Good** - Proves you used your own computer

✓ **Take clear screenshots** - Show Name + Roll No. always

✓ **Write in your own words** - Especially for explanations

✓ **Ask your teacher** - If stuck on something

✓ **Complete on time** - 7 days from assignment given

✓ **Follow instructions** - File names, PDF format, etc.